

Introduction to Software Engineering

Software Testing

The student team is required to complete the **Software Testing** documentation for the assigned course project, following the attached template.



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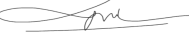
Software Testing

Objectives

This document focus on the following topics:

- ✓ Completing the Software Testing document with the following sections:
 - Test Plan
 - Test Cases
- ✓ Understanding the Software Testing document.

1 Member Contribution Assessment

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2 Test plan

2.1 Introduction

This document presents the Test Plan for the Online Pet Shop System. The purpose of this test plan is to define the testing strategy, testing scope, testing techniques, test objects, and testing activities to ensure that the system satisfies all specified functional and non-functional requirements.

Testing activities described in this document are conducted to detect defects, verify system correctness, and assess overall software quality prior to deployment.

2.2 Test Objectives

The objectives of testing for the Online Pet Shop System are as follows:

- To verify that all system functions operate in accordance with the Software requirements specification.
- To identify defects at early stages of development.
- To ensure data accuracy, consistency, and reliability.
- To validate system security and performance.
- To confirm system readiness for user acceptance.

2.3 Test Objects

2.3.1 *Functional Test Objects*

The following system functions will be tested:

- User registration, login, and logout
- Browsing, searching, and filtering pets.
- Viewing pet details
- Shopping cart management (add, update, remove items)
- Order placement and checkout
- Payment processing
- Order tracking and history
- Administrative functions (product, pet, order, and user management)

2.3.2 *Technical Components*

Testing will be performed on the following technical components:

- Front-end user interface
- Back-end application programming interfaces (APIs)
- Database operations
- Authentication and authorization mechanisms

2.3.3 *Test Documents*

The following project documents will be reviewed and validated:

- Software Requirements Specification (SRS)
- Use Case Diagrams and Use Case Descriptions
- Database design documentation
- User interface specifications

2.4 Test Scope

2.4.1 *In-Scope Testing*

The testing activities included in this plan are:

- Functional testing
- Integration testing
- System testing
- Non-functional testing (performance, security, usability)

2.4.2 *Out-of-Scope Testing*

The following items are excluded from this test plan:

- Internal logic of third-party payment gateways
- External delivery and logistics systems
- Hardware and infrastructure testing

2.5 Testing Techniques

2.5.1 *Black-Box Testing*

Black-box testing techniques will be applied to validate system functionality based on specified requirements without knowledge of internal implementation. The following techniques will be used:

- **Equivalence Partitioning** and **Boundary Value Analysis** for user input validation, particularly in user registration and login forms to verify valid and invalid data ranges.
- **Decision Table Testing** to evaluate system behavior under different combinations of conditions, especially in authentication and checkout processes.
- **Scenario-based Testing** to verify end-to-end user workflows, including searching for pets, adding items to the shopping cart, and completing the checkout process.

Black-box testing will be performed on the following test objects:

- User registration and login modules
- Search and filtering functions

- Shopping cart and checkout workflows

2.5.2 *White-Box Testing*

White-box testing will be conducted to evaluate internal program logic and code structure. The following techniques will be applied:

- **Statement Coverage** to ensure that all executable statements are executed at least once.
- **Branch Coverage** to verify that all decision branches (true and false paths) are properly tested.

White-box testing will focus on the following test objects:

- Backend service functions
- Business logic for order processing
- Price calculation and discount functions

2.5.3 *Experience-Based Testing*

Experience-based testing will be conducted to uncover defects not identified by formal test cases.

Techniques applied:

- Exploratory Testing
- Error Guessing

Test objects:

- User interface interactions
- Error handling and exception flows
- Payment process interruptions

2.6 Testing Levels

2.6.1 *Unit Testing*

Unit testing will be performed by developers to validate individual functions and methods.

2.6.2 *Integration Testing*

Integration testing will verify data flow and interactions among system modules, such as:

- Shopping cart to order processing.
- Order processing to payment handling.

2.6.3 *System Testing*

System testing will evaluate the complete Online Pet Shop System against functional and non-functional requirements.

2.7 Non-Functional Testing

2.7.1 *Performance Testing*

- Page response time
- Concurrent user handling
- Order processing throughput

2.7.2 *Security Testing*

- Authentication and authorization controls
- Password encryption mechanisms

7.3 Usability Testing

- Ease of navigation
- Interface clarity

3 Test cases

3.1 List of test cases

Seq	Test case	Target	Description
1	Valid user registration	User Registration	Register with valid information
2	Invalid email registration	User Registration	Register with invalid email format or existing email
3	Invalid password registration	User Registration	Register with invalid password format
4	Registration with empty required fields	User Registration	Register with empty required fields
5	Successful user login	User Login	Login with valid credentials
6	Login with unapproved account	User Login	Login with account that not being approved by other admin
7	Login with incorrect password	User Login	Login with invalid password
8	Login with unregistered email	User Login	Login with unregistered email
9	Login with empty credentials	User Login	Login with empty email and password
10	Request password reset with registered email	Reset Password	Request OTP using a registered email
11	Request password reset with unregistered email	Reset Password	Request OTP using an unregistered

			email
12	Verify OTP with invalid or expired code	Reset Password	Verify OTP with incorrect or expired OTP
13	Reset password after successful OTP verification	Reset Password	Set a new password after OTP is verified
14	Search with existing keyword	Search	Search with existing keyword
15	Search with non-existing keyword	Search	Search with non-existing keyword
16	Search with empty keyword	Search	Search with empty keyword
17	Filter pets by gender	Filtering	Filter pets by pet's gender
18	Filter pets by colors	Filtering	Filter pets by pet's colors
19	Filter pets by category	Filtering	Filter by category
20	Apply multiple filters	Filtering	Combine multiple filters
21	Add product to shopping cart	Shopping Cart	Add product to cart
22	Add out-of-stock product to cart	Shopping Cart	Add out-of-stock product
23	Update product quantity in cart	Shopping Cart	Update product quantity
24	Update cart quantity to zero	Shopping Cart	Update quantity to zero
25	Remove product from shopping cart	Shopping Cart	Remove product from cart

26	Checkout with complete customer information	Order Processing	Verify order is placed successfully when name, email, phone number, and address are all provided
27	Checkout with missing customer name	Order Processing	Verify order cannot be placed when customer name is empty
28	Checkout with missing customer email	Order Processing	Verify order cannot be placed when customer email is empty
29	Checkout with missing customer phone number	Order Processing	Verify order cannot be placed when customer phone number is empty
30	Checkout with missing customer address	Order Processing	Verify order cannot be placed when customer address is empty
31	Checkout with all required fields empty	Order Processing	Verify order cannot be placed when all required customer information fields are empty
32	Access admin dashboard without authentication	Admin Dashboard	Access admin dashboard page without login

3.2 Test case specifications

3.2.1 Test case Valid user registration

Test case	Valid user registration
Related Use case	User Registration
Context	The user accesses the registration page and enters valid registration information
Input Data	• Name: GiaBao • Email: giabao@gmail.com • Password: Bao@15092005
Expected Output	The system registers the user successfully and displays a confirmation message
Test steps	• Navigate to the registration page • Enter valid name, email, and password • Click the register button
Actual Output	System displays message “Đăng ký thành công” and show the status of account is “Tài khoản đang đợi duyệt”
Result	Passed

3.2.2 Test case Invalid email registration

Test case	Invalid email registration
Related Use case	User Registration
Context	The user enters an invalid email format or an existing email during registration
Input Data	- Name: Gia Bao - Email: giabao12345679 or an existing email - Password: Bao@15092005
Expected Output	The system displays an error message indicating an invalid email format
Test steps	- Navigate to the registration page - Enter valid name, password and invalid email - Click the register button
Actual Output	System displays an error message "Email sai định dạng!" Or "Email đã tồn tại trong hệ thống"
Result	Passed

3.2.3 Test case Invalid password registration

Test case	Invalid password registration
Related Use case	User Registration
Context	The user attempts to register a new account by entering a password that does not meet the system's password requirements.
Input Data	• Name: Gia Bao • Email: giabao@gmail.com • Password: ◦ bao@15092005 ◦ bao15092005 ◦ Bao@05
Expected Output	The system rejects the registration request and displays an error message indicating that the password does not meet the required criteria.
Test steps	• Navigate to the registration page • Enter valid name, email and invalid password • Click the register button
Actual Output	System displays an error message "Mật khẩu phải chứa ít nhất một chữ cái in hoa!" Or "Mật khẩu phải chứa ít nhất 8 ký tự!" Or "Mật khẩu phải chứa ít nhất một ký tự đặc biệt!"
Result	Passed

3.2.4 Test case Successful user login

Test case	Successful user login
Related Use case	User Login
Context	The user enters correct login credentials and the user account has been approved by admin
Input Data	- Email: giabao@gmail.com - Password: Bao@15092005
Expected Output	The system logs the user in and redirects to the admin dashboard
Test steps	- Navigate to the login page - Enter valid credentials - Click the login button
Actual Output	System displays message “Đăng nhập thành công” and redirects to the admin dashboard
Result	Passed

3.2.5 Test case Login with unapproved account

Test case	Login with unapproved account
Related Use case	User Login
Context	The user attempts to log in using an account that has been registered but has not yet been approved or activated by the administrator
Input Data	- Email: giabao@gmail.com - Password: Bao@15092005
Expected Output	The system denies the login request and displays a message indicating that the account is not approved or not yet activated
Test steps	- Navigate to the login page - Enter valid credentials - Click the login button
Actual Output	System displays message “Tài khoản của bạn đang chờ được duyệt. Vui lòng liên hệ quản trị viên.”
Result	Passed

3.2.6 Test case Login with incorrect password

Test case	Login with incorrect password
Related Use case	User Login
Context	The user enters an incorrect password
Input Data	- Email: giabao@gmail.com - Password: a wrong password
Expected Output	The system displays an error message indicating incorrect credentials
Test steps	- Navigate to the login page - Enter valid email and incorrect password - Click the login button
Actual Output	System displays message “Mật khẩu không đúng”
Result	Passed

3.2.7 Test case Reset password with registered email

Test case	Reset password with registered email
Related Use case	Reset password
Context	The user requests a password reset using an email address registered in the system
Input Data	Email: giabao@gmail.com
Expected Output	The system displays a message indicating that password OTP have been sent via email
Test steps	- Open the forgot Password page - Enter a registered email - Click reset password
Actual Output	System display message “OTP đã được gửi đến email của bạn.” and send an email contains OTP for user to verify and change password
Result	Passed

3.2.8 Test case Reset password with unregistered email

Test case	Reset password with unregistered email
Related Use case	Reset password
Context	The user enters an email address that does not exist in the system
Input Data	Email: giabao123@gmail.com
Expected Output	The system displays an error message indicating that the email is not registered
Test steps	- Open the forgot Password page - Enter a registered email - Click reset password
Actual Output	System displays message “Email không tồn tại trong hệ thống”
Result	Passed

3.2.9 Test case Verify OTP with invalid or expired code

<i>Test case</i>	Verify OTP with invalid or expired code
<i>Related Use case</i>	<i>Reset password, Verify OTP sent from email</i>
<i>Context</i>	<i>The user enters an incorrect or expired OTP during the verification process</i>
<i>Input Data</i>	• <i>OTP: 000000</i>
<i>Expected Output</i>	<i>The system rejects the OTP and displays an error message</i>
<i>Test steps</i>	• <i>Enter an invalid or expired OTP</i> • <i>Submit the OTP verification form</i>
<i>Actual Output</i>	<i>System displays message “OTP không hợp lệ.”</i>
<i>Result</i>	<i>Passed</i>

3.2.10 Test case Reset password after successful OTP verification

<i>Test case</i>	Reset password after successful OTP verification
<i>Related Use case</i>	<i>Reset password, Verify OTP sent from email</i>
<i>Context</i>	<i>The user successfully verifies the OTP and sets a new valid password</i>
<i>Input Data</i>	• <i>OTP: Valid OTP</i> • <i>New password: NewPass123</i>
<i>Expected Output</i>	<i>The system resets the password successfully and displays a success message</i>
<i>Test steps</i>	• <i>Enter the correct OTP</i> • <i>Verify OTP successfully</i> • <i>Redirect to reset password page</i> • <i>Enter a new valid password</i> • <i>Submit the reset request</i>
<i>Actual Output</i>	<i>System displays message “OTP hợp lệ.” and redirect to reset password page. After successful reset password the system displays message “Đổi mật khẩu thành công!”</i>
<i>Result</i>	<i>Passed</i>

3.2.11 Test case Search with existing keyword

Test case	Search with existing keyword
Related Use case	<i>Search pets by name or category</i>
Context	<i>The user searches for a product that exists in the system</i>
Input Data	• Keyword: "Corgi"
Expected Output	<i>The system displays a list of matching products</i>
Test steps	• Enter a valid keyword in the search box • Click the search button
Actual Output	<i>System displays list of pet that related to the key word</i>
Result	<i>Passed</i>

3.2.12 Test case Apply multiple filters

Test case	Apply multiple filters
Related Use case	<i>Filter pets</i>
Context	<i>The user applies more than one filter simultaneously to narrow down the list of pets displayed on the product listing page</i>
Input Data	• <i>Category: Dog</i> • <i>Gender: Male</i> • <i>Colors: Yellow</i>
Expected Output	<i>The system displays a list of matching products</i>
Test steps	• <i>Enter a valid keyword in the search box</i> • <i>Click the search button</i>
Actual Output	<i>The system displays only the products that satisfy all selected filter conditions</i>
Result	<i>Passed</i>

3.2.13 Test case Add product to shopping cart

Test case	Add product to shopping cart
Related Use case	<i>Add pet to cart</i>
Context	<i>The user adds an in-stock product to the shopping cart</i>
Input Data	• <i>Product: Available pet item</i> • <i>Quantity: 1</i>
Expected Output	<i>The product is added to the shopping cart successfully</i>
Test steps	• <i>Select a product</i> • <i>Click add to cart</i> • <i>Open the cart page</i>
Actual Output	<i>System displays the shopping cart with the chosen pet</i>
Result	<i>Passed</i>

3.2.14 Test case Update product quantity in cart

Test case	Update product quantity in cart
Related Use case	<i>Update pet quantity in cart</i>
Context	<i>The user updates the quantity of a product in the cart</i>
Input Data	• Updated quantity: 2
Expected Output	<i>The cart updates the product quantity and recalculates the total price</i>
Test steps	• Open shopping cart • Change product quantity • Confirm update
Actual Output	<i>The system updates the pet quantity in the cart and recalculates the total price.</i>
Result	<i>Passed</i>

3.2.15 Test case Checkout with complete customer information

<i>Test case</i>	Checkout with complete customer information
<i>Related Use case</i>	<i>Checkout</i>
<i>Context</i>	<i>The user places an order with all required customer information provided</i>
<i>Input Data</i>	• <i>Name: Dang Beo</i> • <i>Email: dangbeo@gmail.com</i> • <i>Phone: 0123456789</i> • <i>Address: 123 Nguyen Van Cu</i>
<i>Expected Output</i>	<i>The system successfully places the order and displays an order confirmation</i>
<i>Test steps</i>	• <i>Add products to cart</i> • <i>Proceed to checkout</i> • <i>Enter all required information</i> • <i>Choose payment method</i> • <i>Confirm order</i>
<i>Actual Output</i>	<i>System displays message “Đặt hàng thành công!” and display the detail information of the order</i>
<i>Result</i>	<i>Passed</i>

3.2.16 Test case Checkout with all required fields empty

<i>Test case</i>	Checkout with all required fields empty
<i>Related Use case</i>	<i>Checkout</i>
<i>Context</i>	<i>The user attempts to checkout without entering any customer information</i>
<i>Input Data</i>	• <i>Name: empty</i> • <i>Email: empty</i> • <i>Phone: empty</i> • <i>Address: empty</i>
<i>Expected Output</i>	<i>The system blocks checkout and displays validation errors</i>
<i>Test steps</i>	• <i>Proceed to checkout</i> • <i>Leave all required fields empty</i> • <i>Submit the order</i>
<i>Actual Output</i>	<i>System displays error message “Họ và tên không được để trống”; “Số điện thoại không được để trống”; “Email không được để trống”; “Địa chỉ không được để trống”</i>
<i>Result</i>	<i>Passed</i>

3.2.17 Test case Access admin dashboard without authentication

<i>Test case</i>	Access admin dashboard without authentication
<i>Related Use case</i>	<i>Admin Management</i>
<i>Context</i>	<i>A guest user attempts to access the admin dashboard without logging in</i>
<i>Input Data</i>	• <i>Direct access to admin dashboard URL</i>
<i>Expected Output</i>	<i>The system denies access and redirects the user to the login page</i>
<i>Test steps</i>	• <i>Open web</i> • <i>Click admin dashboard directly</i>
<i>Actual Output</i>	<i>The system denies and redirects the user to the login page</i>
<i>Result</i>	<i>Passed</i>